

	1	2	3	4	5	6
A	power					A
B	File: power.kicad_sch					B
C	radio					C
D	File: radio.kicad_sch					D
	1	2	3	4	5	6

wapiti

Sheet: /

File: wapiti.kicad_sch

Title: wapiti trail camera – root

Size: A4

Date: 2026-10-04

Rev: 0.1

KiCad E.D.A. 10.0.5

Id: 1/5

into VSYS_RAW; C1 5F supercap = LTE TX burst + mitten-swap ride-through.

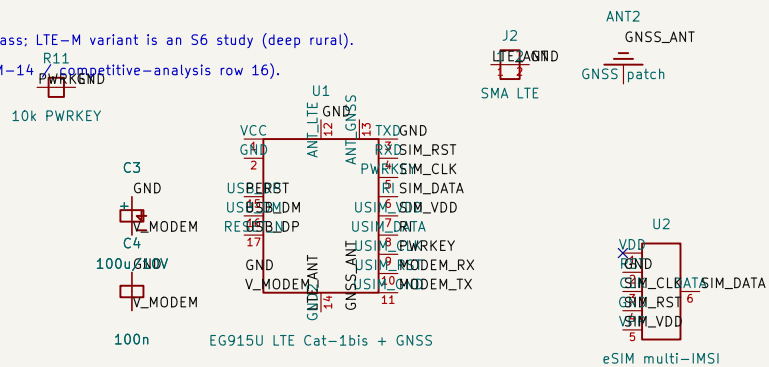


Id: 2/5

5U LTE Cat-1bis + GNSS, eSIM multi-IMSI, SMA LTE antenna

1bis (5M UL) is the photo-delivery class; LTE-M variant is an S6 study (deep rural).

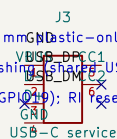
J2 = ^Auser-replaceable SMA (CAM-14 / ^{R11}competitive-analysis row 16).



SS patch ANT2 under the lid RF membrane (0.8 mm GND plastic-only; chassis keep-out).

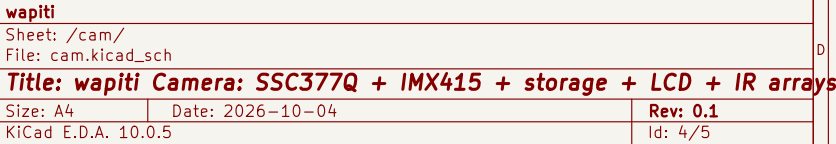
J3 USB-C service: modem firmware + C6 flashing (shared USB_DP/USB_DM).

WAKEUPKEY (10k R11) + RESET_N (PERST from C6 GPIO49); R16 reserved for ring wake.



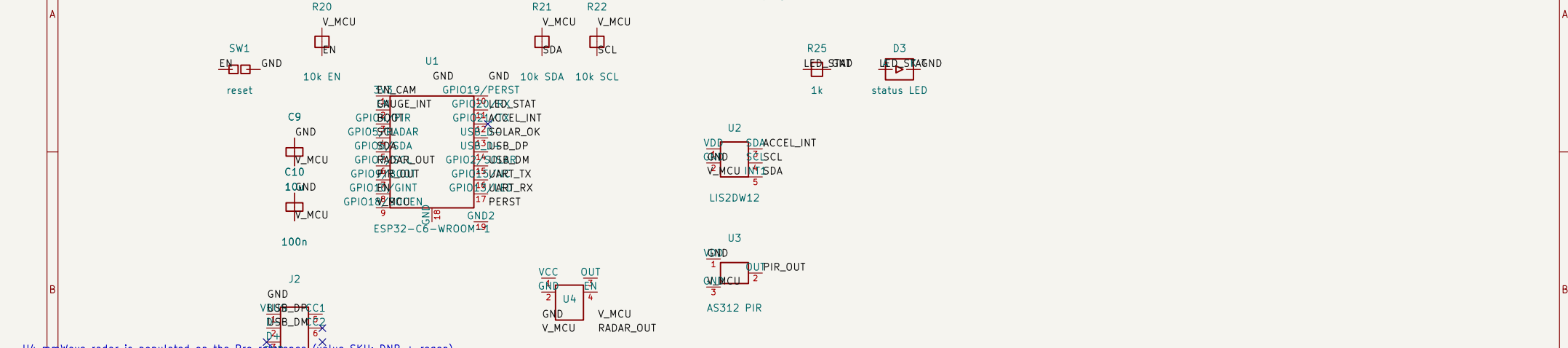
wapiti		
Sheet: /radio/ File: radio.kicad_sch		
Title: wapiti Radio: EG915U LTE Cat-1bis + GNSS + eSIM + SMA		
Size: A4	Date: 2026-10-04	Rev: 0.1
KiCad E.D.A. 10.0.5	Id: 3/5	

IR_940 (no-glow, 80ft) and IR_850 (low-glow, 90ft) are app-selectable (CAM-05).



ESP32-C6 wake supervisor + BLE/Wi-Fi pairing + sensors

C6 owns the heartbeat (HB-01..08): it watches PIR/radar, wakes the SoC rail
(EN_CAM via GPIO18), hosts the fuel gauge on I2C, and gates solar charging
(SOLAR_OK, pack-temp interlock < 0C = PWR-08). BLE pairs the app (PR-03).



U4 mmWave radar is populated on the Pro reference (value SKU: DNP + regen).

J2 USB-C service shares USB_DP/DM with the modem (radio sheet); CC 5.1k on layout.

J1 2x03: 3V3, EN, TX, RX, GND, IO9(BOOT) for factory programming.

wapiti		
Sheet: /mcu/		
File: mcu.kicad_sch		
Title: wapiti MCU: ESP32-C6 supervisor + PIR + radar + accel		
Size: A4	Date: 2026-10-04	Rev: 0.1
KiCad E.D.A. 10.0.5		Id: 5/5